

Lab: Adaptive Control and Parameter Estimation

Objective

Design adaptive control systems that update their parameters online to track changing plant dynamics. You will compare model reference adaptive control (MRAC) with Active Inference parameter learning and analyze stability guarantees.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Mathematical Formulation

Formulate the learning problem mathematically:

1. Define the state-space representation relevant to this module.
2. Specify the cost function or objective for classical approaches.
3. Specify the free energy functional for the Active Inference formulation.
4. Under what assumptions do these formulations yield equivalent solutions?

Write your response here

Part 2: Algorithm Implementation

Design the algorithm in pseudocode:

1. Write the classical learning algorithm step-by-step.
2. Write the Active Inference equivalent step-by-step.
3. Identify shared computational steps (e.g., matrix inversions, prediction steps).
4. Where do the algorithms diverge? What does Active Inference add?

Write your response here

Part 3: Robotic Application

Apply both algorithms to a concrete robotic scenario:

1. Define a specific robot (e.g., 2-DOF arm, mobile robot, quadrotor).
2. Specify numerical parameters (masses, inertias, sensor noise).
3. Trace through 5 timesteps of each algorithm.
4. Compare the resulting state estimates, control actions, or plans.

Write your response here

Part 4: Performance Comparison

Compare the approaches systematically:

Metric	Classical Approach	Active Inference Approach
Computational cost		
Robustness to noise		
Adaptability		
Theoretical guarantees		
Ease of tuning		

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).